4CE

Sovereign *AI Workbench*

Sovereign On-Premise Agentic AI Workbench using Open-Weight Multimodal LLMs for Confidential Industrial Work

Smart India Hackathon 2026 · Problem statement SIH26117

Theme · Smart Automation | Category · Software | Team Kern · 4CE (ACE)

OUR SOLUTION

Four pillars, *one workbench.*

- **Sovereignty**
  Local LLMs → data stays inside
- **Ingestion**
  Legacy files → usable intelligence
- **Governance**
  Tool execution → human approval
- **Trust**
  Generated answer → source-verified



WHY DIFFERENT

Safe by *design.*

Isolated sandbox execution

AI-generated code → isolated environment → controlled execution.

SANDBOXED EXECUTION

```
--network none
--read-only
--cap-drop ALL
```

Exit code: 0

Centralised AI control

Every action → control layer → authorization → execution.

EVERY ACTION, IN ORDER

TONY FRIDAY JARVIS ULTRON YOU

Type approve | Withhold | Approve

Sensitive actions → mandatory human approval

Model-independent architecture

Runs Qwen, Gemma, Mistral or other open-source models.

4CE / TONY (Orchestrator)

- Qwen ✓
- Gemma
- Mistral
- Other open-source models

Change the model, not the architecture.

TECHNICAL APPROACH

Built on an open, *local stack.*

Tech stack

Language & runtime	Python · Bash
Application platform	FastAPI · Uvicorn · Socket.io · SvelteKit · JWT session auth
AI / ML	Bionic · Qwen models · Gemma models · nomic-embed-text-v1.5
Data & storage	SQLite · ChromaDB
Execution & isolation	Docker · Hardening flags
Document generation	python-docx
Deployment & tooling	Custom test harness

PROTOTYPE

75%+

of the prototype is already built

Multi-LLM-Agentic-Ai
GitHub repository · Yash3510

Open the repository

FEASIBILITY · RISK ASSESSMENT & MITIGATION

Every risk, *answered.*

RISK		MITIGATION	
Cloud dependency Data could leave the enterprise	→	Local-only stack + outbound paths disabled	✓
65K context window KV cache exceeded 6 GB VRAM → CPU fallback → 20+ min inference	→	8K context fits VRAM → GPU inference restored	✓
AI hallucination Generated findings could be incorrect	→	ULTRON verification + bounded replan	✓
Generated-code risk AI code could reach the host or network	→	Hardened Docker sandbox → isolated execution	✓

What it takes, *what it pays.*

RESOURCE REQUIREMENTS

- **Local compute + GPU (recommended)**
- **Docker + local model server**
- **Persistent storage + enterprise data**
- **Proven frameworks**

ECONOMIC APPROACH

Lower recurring AI costs

Potentially lower costs for sustained, high-volume workloads

Enterprise licensing

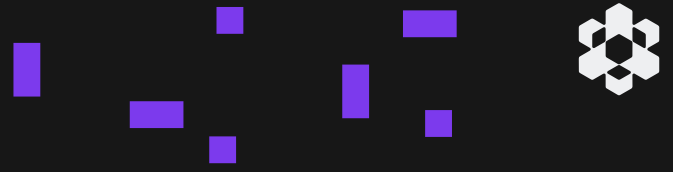
Monetised through licensing plus setup and integration

Recurring revenue

Support, maintenance and custom agents

IMPACT & BENEFITS · A SAFER TOMORROW

Real value, *higher control.*



01

Conversational impact

- Safer AI interactions
- Human oversight

02

Faster research

- Private AI research
- Controlled experimentation

03

National impact

- AI & data sovereignty
- Reduced external dependence

04

Technical impact

- Local LLM + RAG
- Multi-agent orchestration
- Sandbox + verification
- Auditability

05

Economic & industrial impact

- Lower recurring AI costs
- Secure enterprise adoption
- Regulated industries

FUTURE PROSPECTS

From workbench to *control layer.*

01

Multimodal enterprise AI

Vision, audio, video & complex-document understanding

02

Multi-site deployment

Local AI at every site with centralised governance

03

Industry-specific agents

Manufacturing, healthcare, finance, government & defence

04

Continuous security & governance

Policy enforcement, monitoring, audit & compliance

05

Enterprise AI control layer

Evolve into the secure orchestration and governance layer for enterprise AI

RESEARCH & REFERENCES

Grounded in *research.*

SUPPORTING RESEARCH PAPERS

- Attention Is All You Need
- Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks
- Self-Refine: Iterative Refinement with Self-Feedback
- AutoGen: Next-Gen LLM Applications via Multi-Agent Conversation

TECHNICAL DOCUMENTATION

- Sovereign AI Project Deep Research
- Master Architecture + Phase Specifications
- Verification Reports + Security Runbooks
- RUNNING.md + Sovereign Configuration

MARKET RESEARCH

- Inference Economics of Enterprise Coding Agents — 2026
- Hardware Economics: Cloud vs On-Prem TCO — 2026
- On-Premise vs Cloud: Generative AI TCO — 2026

GitHub repo
github.com/Yash3510
Open

Demo video
The 4CE chat, end to end

PPT video
Narrated explanation
Watch

4CE

Your data *stays yours.*

Sovereign. Secure. Verifiable. Actionable.

- Turns private organisational knowledge into actionable intelligence
- Automates real workflows while keeping humans in control

Powerful enough for enterprises. *Controlled enough to trust.*

THANK YOU · TEAM KERN · 4CE (ACE)